1 —

HW #3

a. Prove that $\sum_{n=2}^{\infty} \frac{1}{n(\log(n))^p}$ converges for $p>1$

and diverges for $p \leq 1$

Want to prove using a comparison test. First will prove a lemma: $\sum_{n=1}^{\infty} a_n$ converges iff $\sum_{k=0}^{\infty} 2^k a_{2k}$ converges.

By Theorem 3.24 in Rudin, A series of non-negative terms converges iff its partial sums form a bounded sequence. Therefore we can consider the boundedness of the partial sum

let $s_n = a_1 + a_2 + \cdots + a_n$

$t_n = a_1 + 2a_2 + \cdots + 2^k a_{2k}$

For $n < 2^k$

$$s_n \leq a_1 + (a_2 + a_3) + \cdots + (a_{2^k} + \cdots + a_{2^{k+1}})$$
$$\leq a_1 + 2a_2 + \cdots + 2^k a_{2^k}$$
$$\leq t_k$$

If $n > 2^k$

$$s_n \geq a_1 + a_2 + (a_3 + a_4) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k})$$
$$\geq \tfrac{1}{2} a_1 + a_2 + 2a_4 + \cdots + 2^{k-1} a_{2^k}$$
$$\geq \tfrac{1}{2} t_k$$

$$\Rightarrow \quad 2 s_n \geq t_k$$

Therefore this shows that $\{s_n\}$ and $\{t_k\}$ are either both bounded or both unbounded, proving the lemma

Now $\log n$ is monotonically increasing $\Rightarrow \frac{1}{n \log n}$ decreases,

using the above lemma we want to consider

$\sum 2^k a_{2^k}$ for $\sum \frac{1}{n(\log(n))^p}$

$$\sum 2^k a_{2^k} = \sum_{k=1}^{\infty} 2^k \frac{1}{2^k (\log 2^k)^p} = \sum \frac{1}{(k \log 2)^p} = \frac{1}{(\log 2)^p} \sum_{k=1}^{\infty} \frac{1}{k^p}$$

$\frac{1}{(\log 2)^p}$ is a constant for fixed values of p.

$\sum \frac{1}{k^p}$ converges if $p > 1$ and diverges if $p \leq 1$ as proved in lecture (p-series)

Therefore $\sum \frac{1}{n \log(n)^p}$ converges and diverges similarly, using the comparison with the 2^k lemma.

1b)

Prove that a_n is a decreasing sequence of real numbers and if $\sum a_n$ converges, then $\lim na_n = 0$

$na_n <$ compare to

$$a_n + a_{n-1} + a_{n-2} + \cdots + a_1$$

won't go to 0 as $n \to \infty$

Because always have a_1

$$n/2\, a_n < a_n + \cdots + a_{n/2}$$ for n even, Decreasing Sequence

Know $\sum a_n$ converges $\Rightarrow$ For every $\varepsilon > 0$ $\exists N$ such that $m, n > N$ $|a_n + \cdots + a_m| < \varepsilon$

let $m = n/2$

$\Rightarrow$ $n, n/2 > N$ $\Rightarrow$ $n > 2N$

$$|n/2\, a_n| < |a_n + \cdots + a_{n/2}| < \varepsilon$$

$\Rightarrow$ $n/2\, a_n \longrightarrow 0$

$\Rightarrow$ $n\, a_n \longrightarrow 0$

If n odd, we can modify by adding an additional term: $\Rightarrow$

$$n/2\, a_n < a_n + \cdots + a_{\frac{n+1}{2}}$$

The remainder follows as above.

1c) suppose $\sum a_n$ is convergent $a_n > 0$ $\forall n$

i) does $\sum a_n^p$ converge

WTS $p \geq 1$ converges

$p < 1$ Diverge argue by cases

I. If $p=1$ trivial $\sum a_n^p = \sum a_n \Rightarrow$ converges

II. If $p > 1$ $\sum a_n$ converges $\Rightarrow \lim_{n\to\infty} a_n = 0$

$\Rightarrow$ let $\varepsilon = 1$ $\exists N$ s.t. $n > N \Rightarrow$

$a_n < 1 \Rightarrow$

$a_n^p < a_n < 1$ since $a_n^{p-1} < 1$

$\alpha = p-1 > 0$

$a_n^\alpha < 1^\alpha$ $\alpha > 0$

$\Updownarrow$

$a_n < 1$

III. If $0 < p < 1$ Does not necessarily converge

Counterexample: let $a_n = 1/n^{1/p}$ converges B/C $1/p > 1$

$\Rightarrow \sum a_n^p = \sum 1/n \Rightarrow$ Diverges

IV. $p = 0$ Trivial $\sum a_n^p = \sum 1 \Rightarrow$ Diverge

V. If $p < 0$ Does not necessarily converge

let $a_n = 1/n^2$ $p < 0$

$\sum a_n^p = \sum \left(\frac{1}{n^2}\right)^p = \sum n^{2|p|} \Rightarrow$ Diverges

WTS $\sum \frac{a_n}{1+na_n}$ converges using comparison test.

Given: $a_n > 0$

$\Rightarrow a_n^2 > 0$

$\Rightarrow na_n^2 > 0$

$\Rightarrow na_n^2 + a_n > a_n$

$a_n(1+na_n) > a_n$

$a_n > \frac{a_n}{1+na_n}$

Since $\sum a_n$ converges, the comparison test shows $\sum \frac{a_n}{1+na_n}$ converges

iv) WTS $\sum \frac{a_n}{1+n^2a_n}$ converges using comparison test

Given $a_n > 0$

$a_n^2 > 0$

$n^2a_n^2 > 0$

$a_n + n^2a_n^2 > a_n$

$a_n(1+n^2a_n^2) > a_n$

$a_n > \frac{a_n}{1+n^2a_n^2}$

$\Rightarrow$ Converges by Comparison Test.

vi. $\sum \frac{\sqrt{a_n}}{n}$ $\sum a_n$ converges and $a_n > 0$ $\forall n$

<u>Lemma:</u> WTS $\sum \sqrt{a_n \cdot b_n}$ converges if $\sum a_n$ converges and $\sum b_n$ converges and $a_n, b_n > 0$ $\forall n$

If we let $a_n = a_n$ $b_n = 1/n^2$ then we can prove

$\sum \frac{\sqrt{a_n}}{n}$ converges b/c $\sqrt{a_n \cdot b_n} = \frac{\sqrt{a_n}}{n}$

If $a_n > 0$ & $b_n > 0$ $\Rightarrow$ $a_n^2 > 0$ $b_n^2 > 0$ trivially

$$a_n b_n < \tfrac{1}{2} a_n^2 + \tfrac{1}{2} b_n^2 + a_n b_n$$
$$a_n b_n < \tfrac{1}{2}(a_n^2 + b_n^2 + 2 a_n b_n)$$
$$a_n b_n < \left(\tfrac{1}{\sqrt{2}}(a_n + b_n)\right)^2$$
$$\sqrt{a_n b_n} < \tfrac{1}{\sqrt{2}}(a_n + b_n)$$

Right side converges ~~because~~ using limit laws, ie. constant times the sum of two convergent series converges.

$\Rightarrow$ $\sum \sqrt{a_n b_n}$ converges by comparison test

$\Rightarrow$ $\sum \frac{\sqrt{a_n}}{n}$ converges using lemma $\sum \sqrt{a_n b_n}$

$a_n = a_n$ convergent given

$b_n = 1/n^2$ convergent p-series

We an example of a convergent series $\sum a_n$ for which $\sum a_n^2$ diverges.

Choose $a_n = \frac{(-1)^n}{\sqrt{n}}$

By the alternating series theorem this converges.

If we consider $\sum (-1)^n a_n \Rightarrow a_n$ is a decreasing sequence, therefore $a_1 \geq a_2 \geq \cdots \geq a_n \cdots \geq 0$ and $\lim a_n = 0$. as proved in class $\lim a_n = 1/n^p = 0$ at $p = 1/2 > 0$

Now $\sum a_n^2$ diverges

$$a_n^2 = \left[\frac{(-1)^n}{\sqrt{n}}\right]^2 = \frac{(-1)^{2n}}{n} = \frac{1}{n}$$

$\Rightarrow$ and we know $\sum 1/n$ harmonic series diverges